



Intelligent Control of Double Boost Converter Interfaced with Multilevel Inverter for Electrical Vehicle Applications

Praveen Kumar Varanasi¹ · S. Nagaraja Rao¹ · Punithavathi Duraiswamy²

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Abstract This work presents a double boost converter (DBC) integrated with three-level diode clamped multilevel inverter (TLDCI). The proposed converter can be used for renewable energy, grid or electric vehicle applications. The double boost converter provides twice the output voltage in comparison with convention boost converter. The DBC is modelled and simulated under open loop and closed loop operating conditions. Proportional integral (PI) and fuzzy logic controllers (FLCs) are used in closed loop operation. The DBC further integrated with three-level diode clamped inverter. Sine and third harmonic pulse width modulation (SPWM and TH-PWM) is used to generate pulses for TLDCI. The proposed converter simulated in MATLAB/Simulink. Comparison results in open and closed loop such as output voltage error for double boost converter and line, phase voltages and total harmonic distortion for inverter output voltage and current are presented.

Keywords DBC · TLDCI · FLC · SPWM · TH-PWM · THD

Introduction

The high frequency dc–dc converters provide optimum solution for energy conversion from renewable energy sources. There are several topologies in DC–DC converter which

serve the purpose [1]. Achieving higher level output with minimum components and less control complexity is one of the most promising area of interest for researchers [2]. Numerous control methods are employed to govern DC–DC converters, but those with a simple and resilient design, low cost, and good performance under all conditions are of particular interest. Conventional controllers like P, PI, and PID linear control techniques are inaccurate when significant changes in load parameters are present [3, 4].

Intelligent control techniques like fuzzy logic able to provide optimal solutions for large spectrum of engineering problems, including dc to dc converters [5, 6]. Fuzzy control is popular, since fuzzy control's membership and if–then principles represent how humans approach a control problem organically. Fuzzy control is clearly beneficial in situations when the system is hard to model due to non-linearity, imprecision and complexity [7, 8].

The present work mainly focuses on achieving the output voltage (fourfold) with minimum number of components by proposed double boost converter (DBC) [9, 10]. The output of the DBC is connected with three level inverter. The AC output from the inverter can be used to supply grid or electric vehicles [11, 12]. The control algorithms provided in the present work provide most accurate results at the output.

The similar results can be achieved by implementing Z-Source Inverter (ZSI) in the place of a boost converter. The conventional ZSI has a cross-legged impedance network followed by an inverter. This converter has the advantage of accommodating shoot through states in its operation. Hence, the ZSI can harvest an output voltage greater than the dc bus voltage available. The inverter's durability has also been considerably increased shot throughs. Impedance source inverters have disadvantages of higher THD levels and discontinuous input currents, so for the current research, high gain boost converter with MLI has been adopted [13, 14].

✉ S. Nagaraja Rao
nagarajaraao.ee.et@msruas.ac.in

¹ Department of Electrical Engineering, M.S. Ramaiah University of Applied Sciences, Bangalore 560054, India

² Department of Electronics and Communication Engineering, M.S. Ramaiah University of Applied Sciences, Bangalore 560054, India

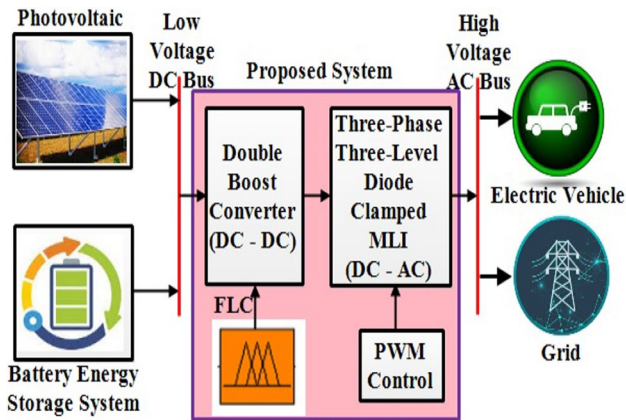


Fig. 1 Proposed DBC interfaced with TLDCI for RES and EV applications

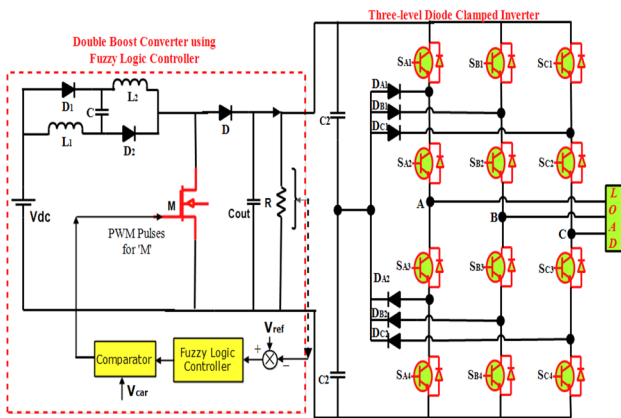


Fig. 2 Proposed DBC converter with three-level TLDCI circuit configuration

The block diagram and circuit topology of proposed double boost converter (DBC) integrated with three level diode clamped multilevel inverter (TLDCI) are shown in Figs. 1 and 2, respectively. The converter has two stages of power conversion. In stage -1, a double boost converter processes the input to a higher level at its output; in stage-2, the three level diode clamped multilevel inverter provides a three phase output voltage at the load [15, 16].

Operation of DBC

The topology of DBC is shown in Fig. 2. It operates in two modes. In mode 1, the switch M is closed, and in mode 2, switch M is opened. Mode 1 of operation is similar to boost mode in conventional boost converter. The inductors L_1 , L_2 and capacitor C_1 store energy. When switch is opened, i.e., in mode 2, the stored energy will be released to load. Figures 3 and 4 give the equivalent circuits for mode 1 and 2, respectively.

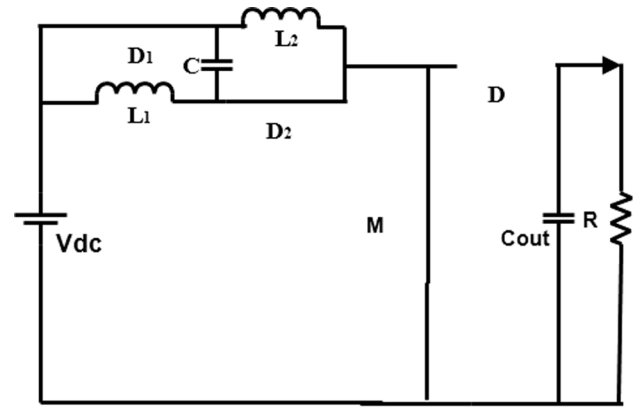


Fig. 3 Mode-1 operation of DBC

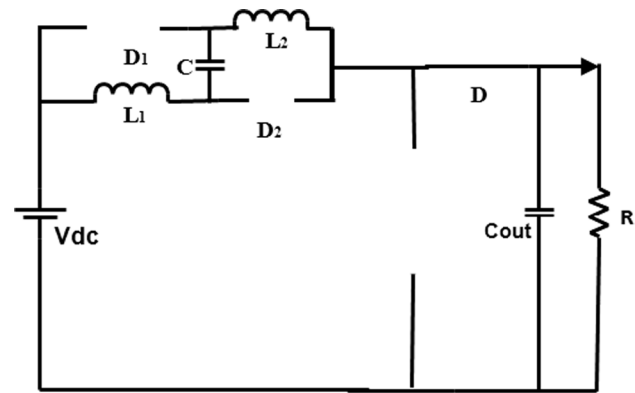


Fig. 4 Mode-2 operation of DBC

In mode 1 of operation

$$V_{dc} = V_{L1} = V_{L2} = V_C \quad (1)$$

In mode 2 of operation

$$V_0 = V_{dc} + V_{L1} + V_{L2} + V_C \quad (2)$$

But

$$V_C = V_{dc} \quad (3)$$

$$V_{L1} = V_{L2} = V_L \quad (4)$$

Substituting (3) and (4) in (2) gives

$$2V_L = V_0 - 2V_{dc} \quad (5)$$

$$V_L = \frac{V_0 - 2V_{dc}}{2} \quad (6)$$

Applying volt-sec balance principle

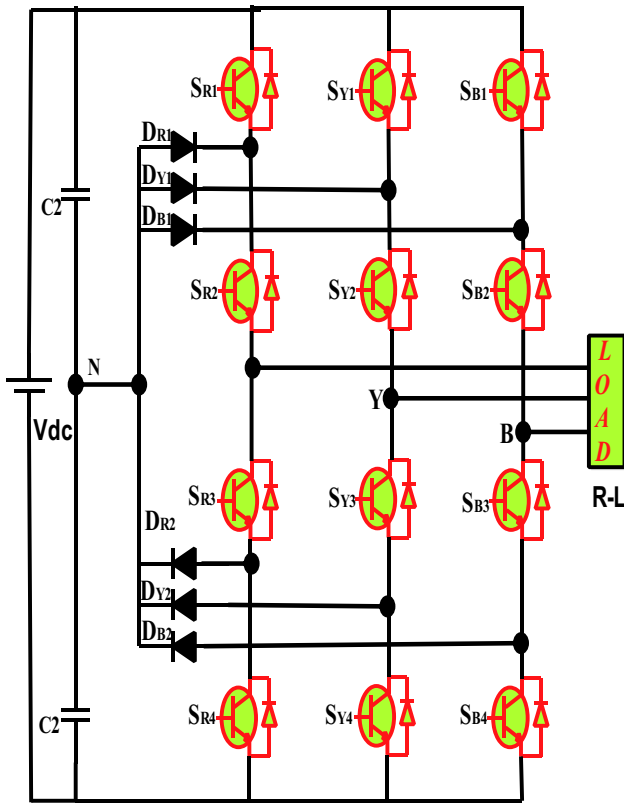


Fig. 5 Topology of three phase TLDCI

$$DTV_{dc} = \left(\frac{V_0 - 2V_{dc}}{2} \right) (1 - D)T \quad (7)$$

Rewriting (7)

$$\frac{V_0}{V_{dc}} = \left(\frac{2}{1 - D} \right) \quad (8)$$

Close observation of Eq. (8) reveals that the output of conventional boost converter should multiply by 2 in order to obtain the output of DBC, and hence, the name double boost converter acquainted.

Topology of three phase TLDCI

Figure 5 depicts the topology of three phase TLDCI. This topology needs ‘6(m-1)’ power switches, ‘(m-1)’ DC bus capacitors, ‘6(m-1)’ main diodes, ‘3(m-1)*(m-2)’ clamping diodes for ‘m’ level output. The operation of TLDCI can be explained by considering the switching sequence shown in Table 1 [17].

Table 1 TL-DCI switching sequence of Phase ‘R’

Sl. No	ON switches in phase ‘R’	Diodes in Phase ‘R’	Voltage
1	$S_{A1}, S_{A2} / S_{B1}, S_{B2} / S_{C1}, S_{C2}$	—	$+V_{dc}/2$
2	$S_{A2}, S_{A3} / S_{B2}, S_{B3} / S_{C2}, S_{C3}$	$D_{A1}, D_{A2} / D_{B1}, D_{B2} / D_{C1}, D_{C2}$	0
3	$S_{A3}, S_{A4} / S_{B3}, S_{B4} / S_{C3}, S_{C4}$	—	$-V_{dc}/2$

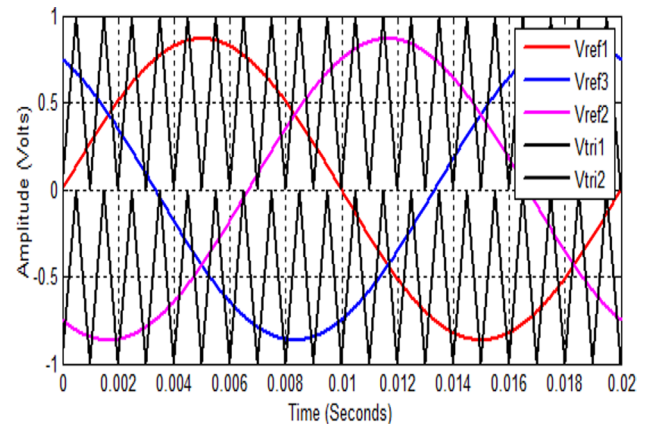


Fig. 6 Sine reference-based PWM technique

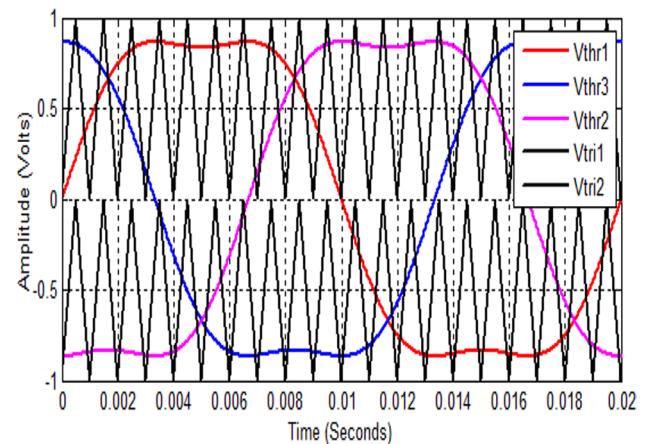


Fig. 7 Third harmonic-based PWM technique

Control of TLDCI

The proposed TLDCI is controlled by using sine and third harmonic injection PWM techniques. The carrier and reference waves in sine and third harmonic techniques are shown in Figs. 6 and 7, respectively. The equations governing the aforementioned techniques are given (9) through (12). Equations (10), (11) and (12) represent the carrier waves in three phases each phase shifted by 120° [18, 19].

$$V_{\text{Sin}} = \frac{\sqrt{3}}{2} \sin \theta \quad (9)$$

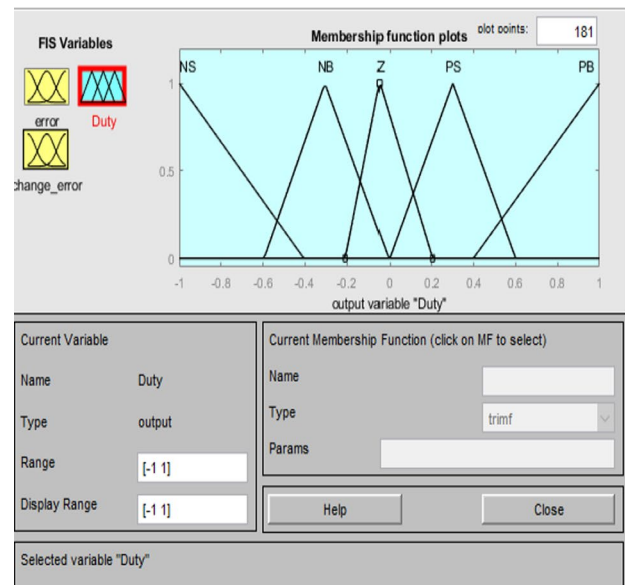
$$V_{\text{thi1}} = \frac{2}{3\sqrt{3}} \sin \theta + \frac{1}{3\sqrt{3}} \sin 3\theta \quad (10)$$

$$V_{\text{thi2}} = \frac{2}{3\sqrt{3}} \sin(\theta - 120) + \frac{1}{3\sqrt{3}} \sin(3\theta - 120) \quad (11)$$

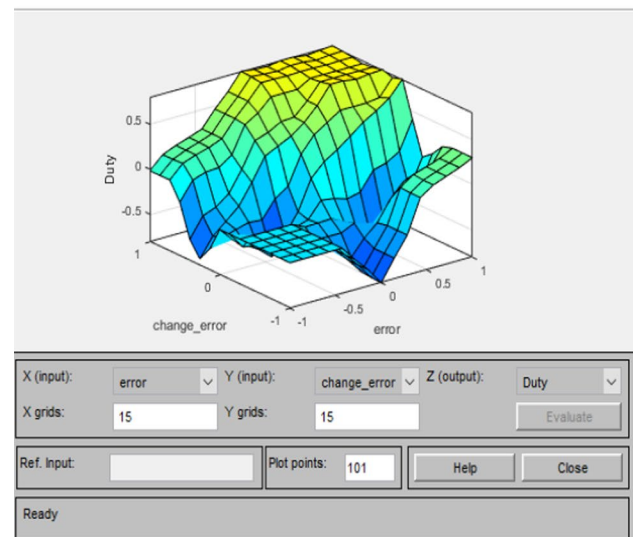
$$V_{\text{thi3}} = \frac{2}{3\sqrt{3}} \sin(\theta - 240) + \frac{1}{3\sqrt{3}} \sin(3\theta - 240) \quad (12)$$

Control of DBC

Amongst digital controllers, FLC is most prominent controller and works based on set of rules extracted from system behaviour and knowledge base. FLC improvises dynamic response even for nonlinear systems, which is not achievable with traditional controllers. Fuzzification, interface mechanism, rule basis, and defuzzification are the four main components of the FLC process. In the present work, FLC provides the duty cycle for output voltage control of the DBC. Error and change in error were obtained by comparing the obtained output with reference voltage. The error and change in error signal are alienated into five subsets known as NB, NS, PS, and PB, fuzzy rule base are given in Table 2. FLC implementation process with duty cycle generation and rule surface are characterized in Fig. 8a, b, respectively. Fuzzification converts the input values from crisp to fuzzy, maps the input values to fuzzy membership functions, and assesses the degree of membership of each input in each membership function. Membership function gives graphical interpretation for each input to represent its degree of participation in a set. Triangular membership function is used in this work characterised by three parameters, and the membership function $\mu(x)$ is described as illustrated



(a)



(b)

Fig. 8 a Duty cycle generation b rule surface

Table 2 Parameters

Parameters	Value/range
Input voltage of DBC	50 V
MOSFET operating frequency	25 kHz
L_1 and L_2 of DBC	10 mH
C_1 , C_2 and C_{out} of DBC	47 μF and 470 μF
Input voltage applied to TLDCI	200 V
C_1 and C_2 of TLDCI	1 μF
Range of modulation index	0 to 1
Load	$R = 10 \Omega$ and $L = 20 \text{ mH}$

in Fig. 8a. These membership function values are derived from expert knowledge [20–22].

Results and Discussion

The simulation analysis of DBC interfaced with three-level DCI was presented. R-L load is considered for TLDCI. The DBC simulation results in open loop mode and in closed loop with PI control and FLC are presented. The parameters of DBC interfaced with three-level DCI are listed in Table 2. The DBC operates at a frequency of 25 kHz.

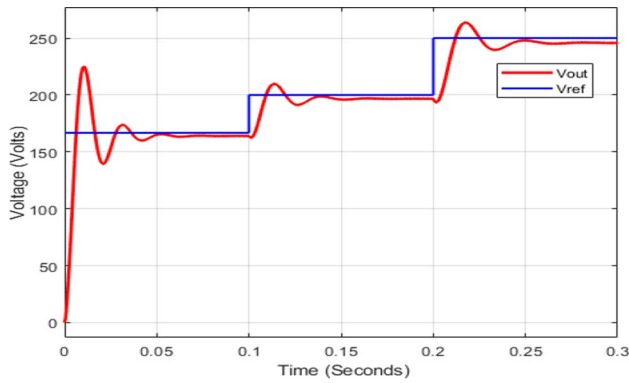


Fig. 9 Output of DBC using open loop control for various duty ratios

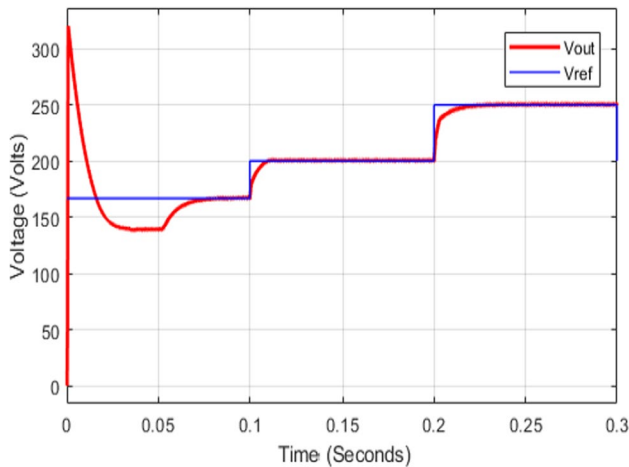


Fig. 10 Output of DBC using PI control for various duty ratios

Figure 9 depicts the DBC output in an open loop mode for a variable duty ratio of 0.4, 0.5, and 0.6, with simulated output voltages (V_{out}) of 163.27 V, 197.57 V, and 246.8 V, respectively. However, according to Eq. (8), the actual theoretical values (V_{ref}) should be 166.66 V, 200 V, and 250 V for duty values 0.4, 0.5, and 0.6, respectively. This means that the output voltage will have wide variations from the input voltage.

Figure 10 illustrates an output from a DBC controlled by PI control, with a changeable reference voltage (V_{ref}) of 166.66 V, 200 V, and 250 V by considering 0.4, 0.5, and 0.6 duty cycles. The respective ' V_{out} ' values of DBC are 166.21 V, 199.62 V and 249.56 V. From Fig. 9, it can be noted that the output of DBC is automatically changed in accordance with ' V_{ref} ' values.

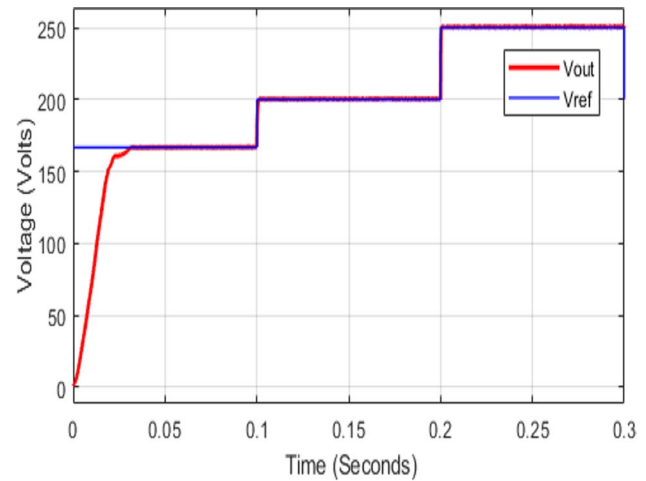


Fig. 11 Output of DBC using FLC for various duty ratios

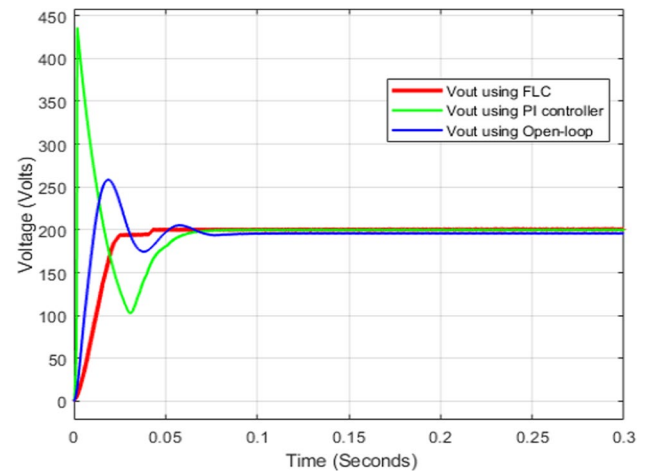


Fig. 12 Output of DBC using open loop, PI and fuzzy logic controllers for fixed duty ratio

Figure 11 illustrates an output from a DBC controlled by intelligent control method, i.e., FLC control, with a changeable reference voltage (V_{ref}) of 166.66 V, 200 V, and 250 V by considering 0.4, 0.5, and 0.6 duty cycles. The respective ' V_{out} ' values of DBC are 166.65 V, 199.98 V and 249.9 V. From Fig. 11, it can be noted that the output of DBC is automatically changed in accordance with ' V_{ref} ' values (Figs. 12, 13, 14, 15, 16, 17).

Figures 18, 19 and 20 give the THD percentages under sine and third harmonic PWM techniques. Line current THD percentages under SPWM and TH-PWM are 2.23% and 2.27%, respectively.

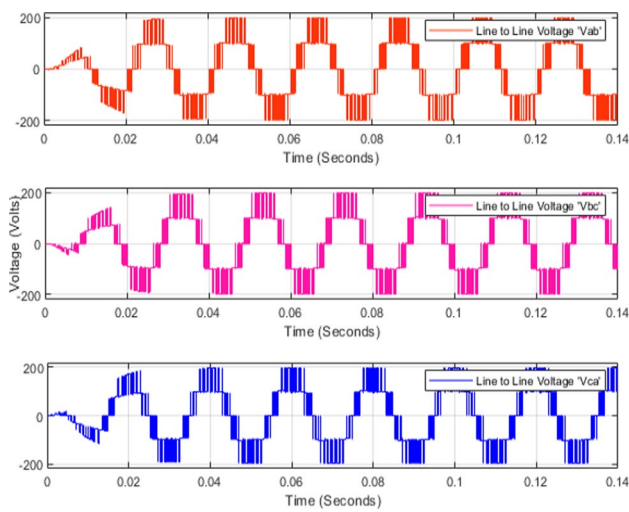


Fig. 13 Line-to-line output voltages of TLDCI

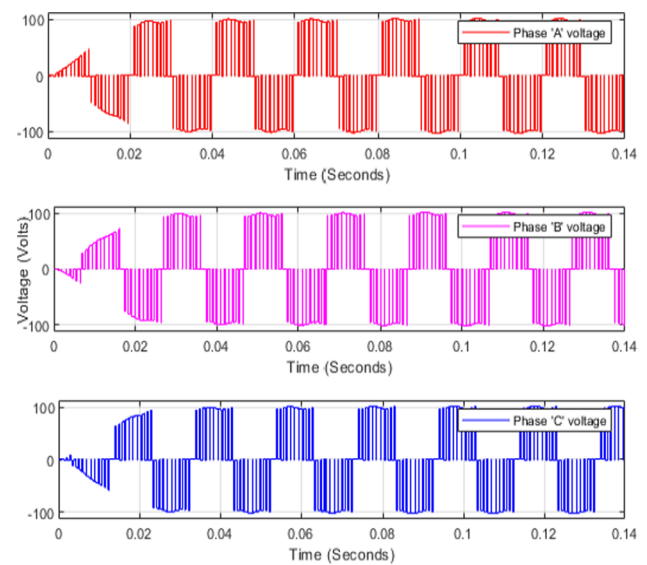


Fig. 15 Phase output voltages of TLDCI

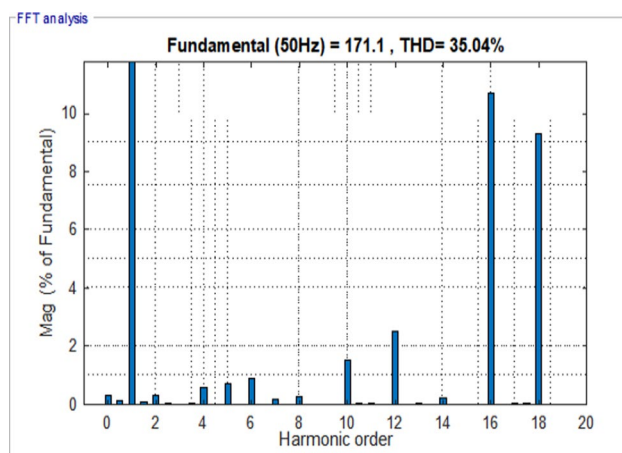


Fig. 14 Line-to-line output voltage THD of TLDCI

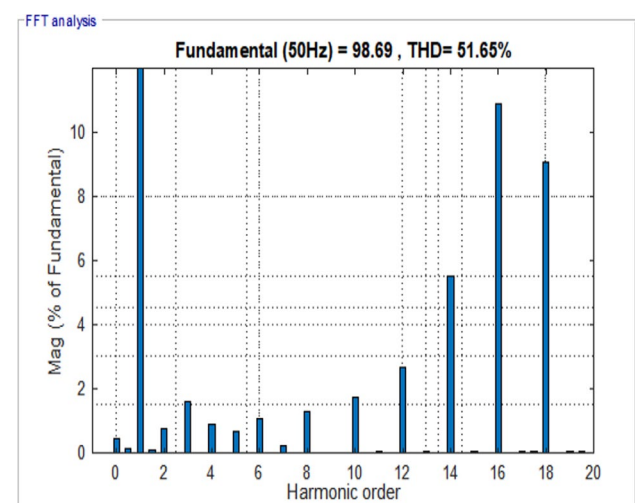


Fig. 16 Phase output voltage THD of TLDCI using Sine PWM

Tables 3, 4 and 5 give the gist of the simulation results presented in Figs. 8, 9 and 10. Parameters such as rise time, peak time, peak overshoot and settling times are compared in both open loop and closed loop conditions. The last column in the three Tables 3, 4 and 5 gives the voltage deviation

from the actual or expected output. It is observed that voltage error (difference of actual and obtained) is quite low for fuzzy logic-based feedback controller.

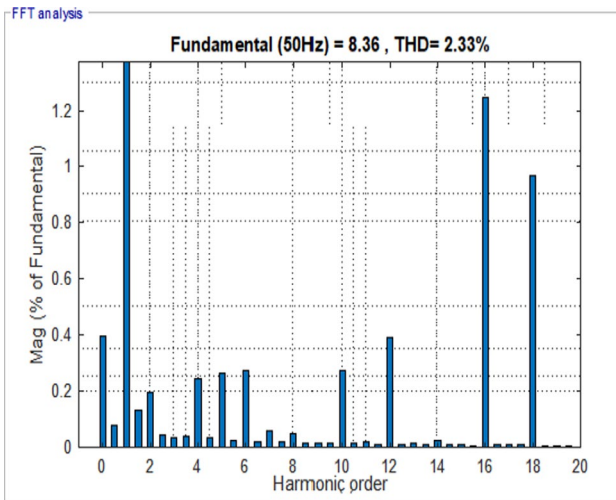


Fig. 17 Line current output THD of TLDCI

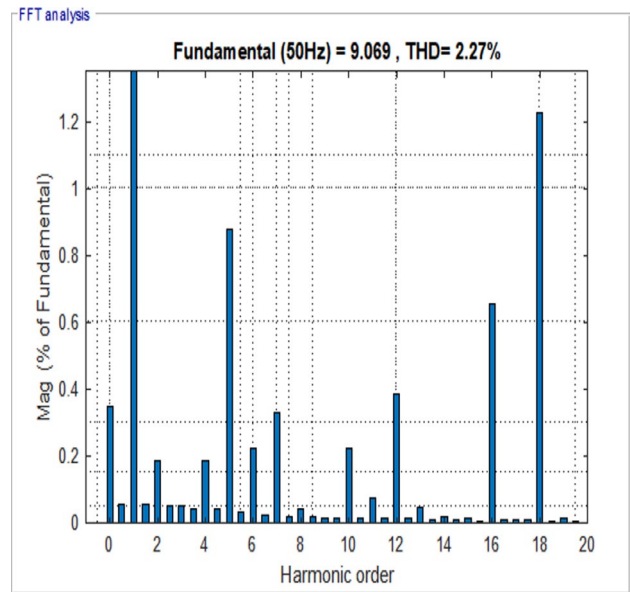


Fig. 19 Phase output voltage THD of TLDCI using TH- PWM

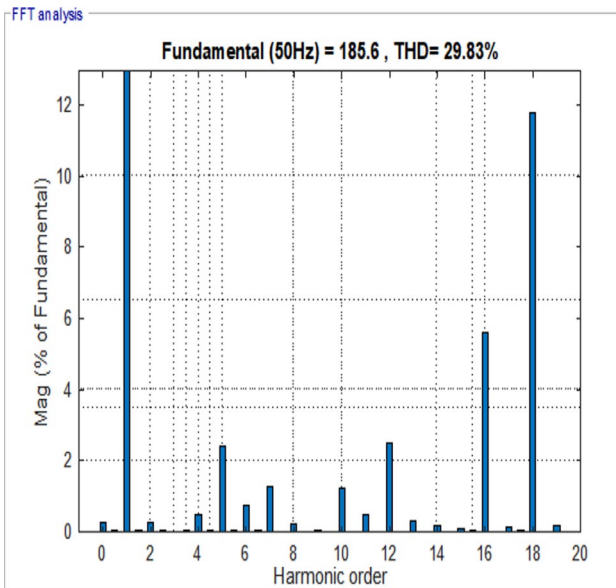


Fig. 18 Line-to-line output voltage THD of TLDCI using TH -PWM

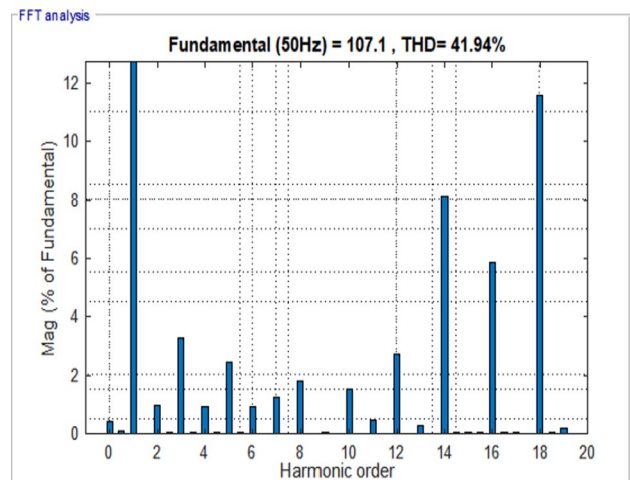


Fig. 20 Line current output THD of TLDCI using TH-PWM

Table 3 Simulation readings DBC using open loop

S.No	V_{dc} (V)	Expected voltage V_{ref} (V)	Rise time (t_s in mS)	Peak time (t_p mS)	Peak overshoot (M_p in %)	Settling time (t_s mS)	V_{out}	Voltage deviation (V)
1	50	166.66	9.315	10.35	30.09	81.15	163.27	3.39
2	50	200	11.34	12.6	28.15	117.2	197.57	2.43
3	50	250	14.07	16.23	24.52	29.76	246.8	3.2

Table 4 Simulation readings of DBC with PI control

S.No	V _{dc} (V)	V _{ref} (V)	Rise Time (t _s in mS)	Peak Time (t _p mS)	Peak Over-shoot (M _p in %)	Settling Time (t _s mS)	V _{out}	Voltage deviation (V)
1	50	166.66	7.3	0.1	60.81	90	166.21	0.45
2	50	200	8.5	0.1	55.23	85	199.62	0.38
3	50	250	9.2	0.1	48.11	81	249.56	0.44

Table 5 Simulation readings of DBC with FLC

S. no	V _{dc} (V)	V _{ref} (V)	Rise time (t _s in mS)	Peak Time and overshoot	Settling Time (t _s mS))	V _{out}	Voltage deviation (V)
1	50	166.66	21.77	-	83.28	166.65	0.035
2	50	200	25.97	-	86	199.98	0.02
3	50	250	30.90	-	100	249.9	0.1

Conclusion

The DBC integrated with three phase TLDCI for grid or electrical vehicle applications is presented. The proposed DBC is simulated in open loop and closed loop. For closed loop operation, PI and fuzzy logic-based controllers were used. The results in aforementioned conditions were compared. It is observed that the DBC operating in closed loop with fuzzy logic controller has no overshoot and less rising and settling times in comparison with PI controller. With the PI controller, the DBC producing the output voltage deviations in the range of 380 mV–450 mV. From Table 3, it is evident that DBC with FLC has very less voltage deviation in the order or 35 mV–100 mV for three different boost factors. Hence, it can be concluded that the optimum results can be obtained by employing fuzzy logic control for the proposed DBC. The three phase TLDCI is simulated by considering SPWM and TH-PWM control techniques. Both the control techniques were analysed by comparing the THD levels for both output voltage and current waveforms. The THD levels obtained by TH-PWM are very less.

Future Scope

The proposed DBC integrated with three phase TLDCI can be implemented with a space vector modulation technique and extended to higher levels. Also, the proposed DBC integrated with three phase TLDCI can be interfaced with brushless DC motor to drive the wheels of electrical vehicles.

Declarations

Conflicts of interest On behalf of all the authors, the corresponding author states that there is no conflict of interest.

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